Multilingual Emotion Detection

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Introduction

- Classifier detecting six key emotions in sentences.
- Fine tuned model on three languages: English, German, and Algerian Arabic.
- Compare performance on low resource and unseen languages: Persian, Spanish, Hausa, and Amharic using 50 translated sentences.
- Address low-resource and imbalanced labels.

Motivation

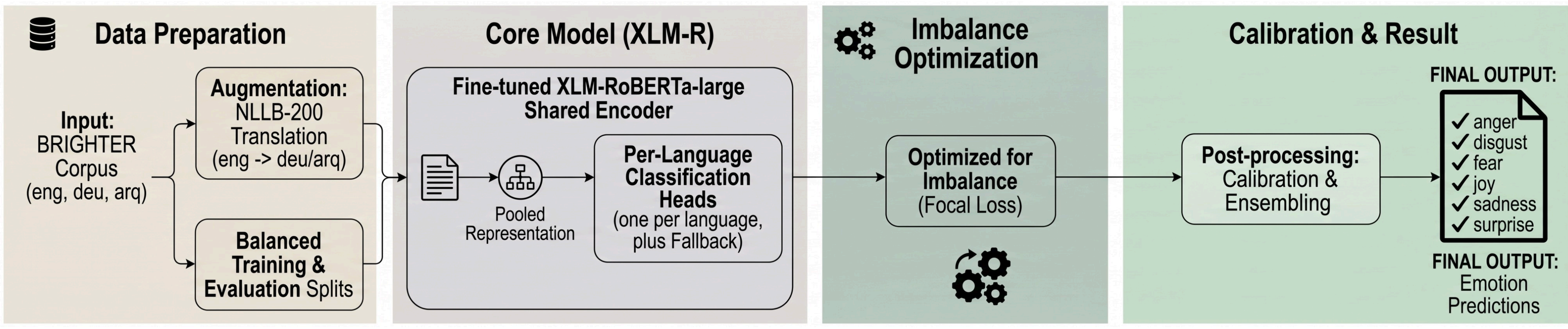
- Evaluate model performance across High Resource vs Low Resource languages.

Research Question

- Can multilingual models detect emotions in both trained and unseen languages?
- Can focal loss, augmentation, and calibration mitigate training data label imbalance?

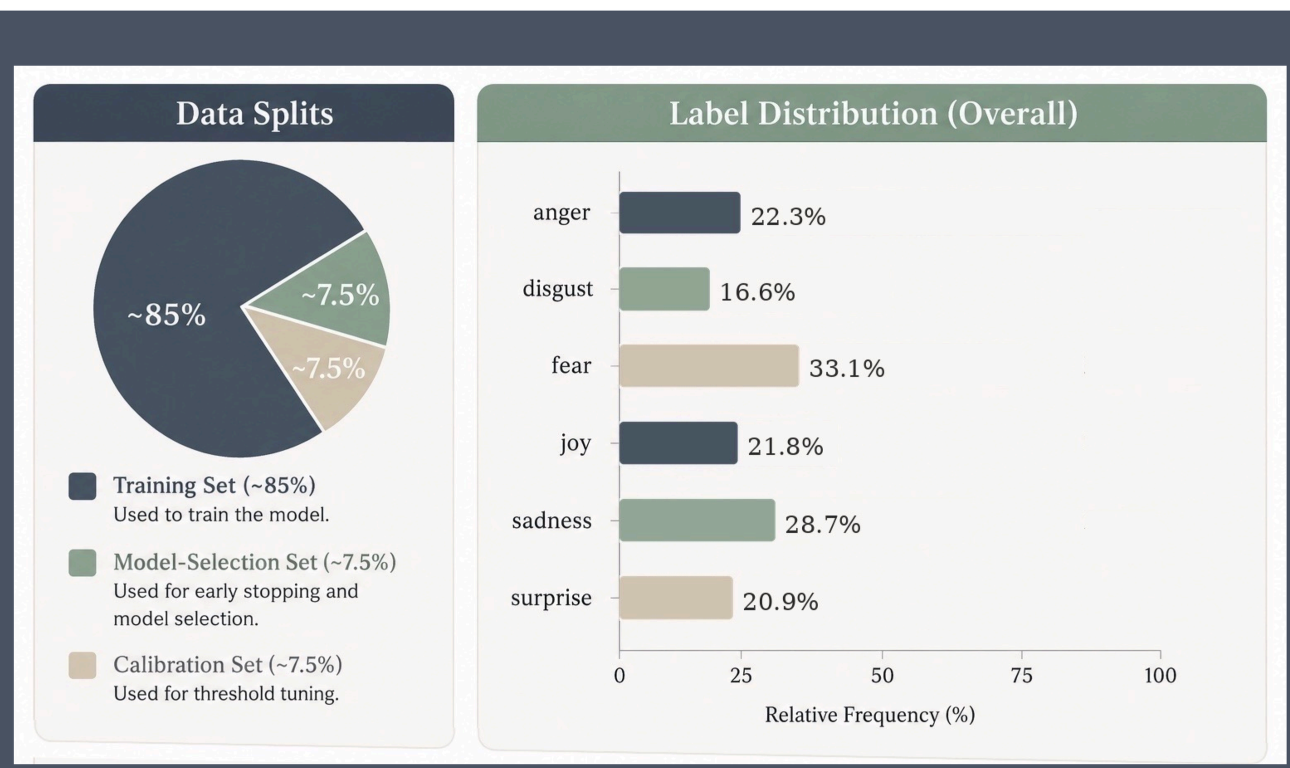
Methodology

- Fine-tune XLM-RoBERTa-large for multi-label emotion prediction with per-language classification heads.
- Handle label imbalance using focal loss, per-label positive weights, and language-balanced sampling.
- Improve low-resource and final predictions using translated silver data, threshold calibration, and ensembling.



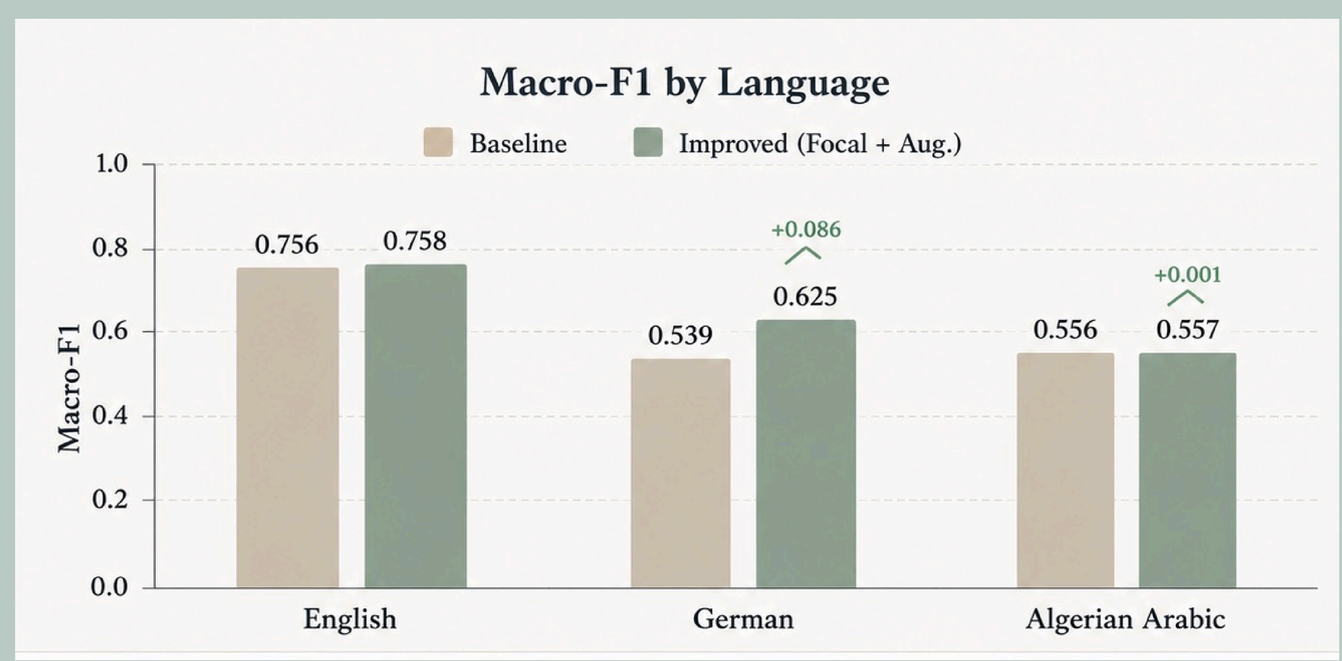
Data

- Dataset: BRIGHTER corpus from SemEval-2025 Task 11.
- Data type: short multilingual texts.
- Languages: English, German (High Resource), and Algerian Arabic (Low Resource).
- Labels: anger, disgust, fear, joy, sadness, surprise (English excludes disgust in scoring)
- Task format: multi-label classification, each text can have multiple emotions.
- The official dev set was held out and scored once using per-language macro-F1.



Results & Evaluation Scores

- The overall score increased from 0.617 to 0.647 with the improved method.
- The largest gain was observed for German, from 0.539 to 0.625.
- English performed best, German second, and Algerian Arabic the worst.
- Anger and fear were easiest to predict; joy and surprise were hardest.
- These techniques mainly improved rare German emotion labels.
- The improved XLM-RoBERTa-large model achieved the best result: 0.647 macro-F1.

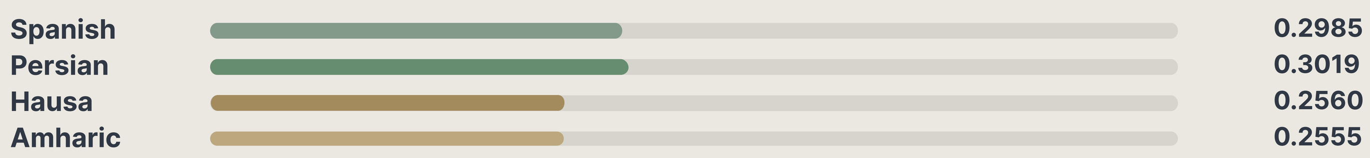


Conclusions

- Multilingual emotion detection remains challenging for low-resource languages.
- Handling rare emotion labels is key for stronger cross-lingual performance.
- Language-specific design choices improved model reliability.
- Zero-shot tests even on high-resource languages perform worse than low-resource seen languages.
- Multilingual transfer works, but still depends on data quality and label balance.

Unseen Language Evaluation

- Macro-F1 scores were measured on unseen languages: Spanish, Persian, Hausa and Amharic on 50 sentences.



References

- BRIGHTER Corpus / SemEval-2025 Task 11
- XLM-RoBERTa-large
- NLLB-200 Machine Translation
- Focal Loss for Class Imbalance